

CLARKSVILLE OUTLAW, TN, USA

WMO: 723280

Lat: 36.624N

Lon: 87.419W

Elev: 560

StdP: 14.40

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03894

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.1	17.3	1.0	5.9	17.2	5.8	7.6	19.8	24.6	41.7	22.2	42.3	6.7	340	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	93.5	74.9	91.3	75.1	89.7	74.6	78.3	88.8	77.4	87.6	76.4	85.8	7.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	135.2	84.1	74.3	130.6	83.0	73.2	126.1	81.8	42.2	89.1	41.1	87.6	40.1	86.0	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.1	17.8	15.8	6.4	97.4	6.1	4.8	1.9	100.9	-1.6	103.7	-5.1	106.5	-9.6	110.0	
			DB												
			WB	5.2	80.3	5.7	1.7	1.1	81.6	-2.2	82.6	-5.4	83.5	-9.5	84.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.9	37.5	40.3	49.4	59.0	67.8	75.8	77.9	77.5	71.4	59.9	48.1	40.8
	DBStd	16.99	11.98	11.71	10.86	8.92	7.19	4.70	4.41	4.80	6.34	8.68	9.30	10.33
	HDD50	1381	415	313	149	25	0	0	0	0	15	142	322	
	HDD65	3802	852	693	493	217	60	1	0	1	18	206	510	751
	CDD50	4624	28	42	129	294	553	774	866	852	643	322	85	36
	CDD65	1570	0	1	9	36	148	325	401	387	211	48	3	1
	CDH74	15150	0	10	76	358	1269	3224	3981	3791	1936	486	19	0
	CDH80	5988	0	2	7	47	348	1303	1706	1697	753	123	2	0
	WSAvg	6.3	8.2	8.1	8.1	7.6	5.9	5.2	4.7	4.3	4.6	5.7	6.5	7.2
	PrecAvg	48.7	3.7	4.0	5.0	4.2	4.6	4.1	3.9	3.5	3.3	3.1	4.4	4.8
Precipitation	PrecMax	63.1	9.7	10.8	15.9	9.0	11.6	11.1	9.0	9.0	11.6	7.7	8.9	12.5
	PrecMin	32.2	0.8	1.2	1.8	0.8	0.8	0.5	1.1	0.6	0.4	0.0	1.3	1.0
	PrecStd	7.4	2.0	2.1	2.7	1.9	2.2	2.2	1.8	2.0	2.2	1.8	1.9	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	68.1	73.7	80.8	84.2	88.6	94.9	96.2	99.6	94.3	88.0	76.5	71.4	
	0.4% MCWB	60.2	61.0	64.0	66.0	71.6	72.2	76.5	74.1	72.6	69.7	62.6	63.4	
	2% DB	64.1	68.6	75.6	81.2	86.1	91.1	92.5	94.7	90.8	83.2	72.0	66.1	
	2% MCWB	57.4	60.0	61.9	64.2	70.5	74.8	76.6	74.5	72.3	67.7	60.5	59.7	
	5% DB	61.2	64.3	71.7	78.5	84.0	89.5	90.6	91.2	88.0	79.6	67.6	61.8	
	5% MCWB	55.5	55.9	59.6	63.2	69.3	74.6	76.2	75.1	72.4	65.3	58.2	56.5	
	10% DB	57.1	59.4	67.3	75.1	81.6	87.5	88.6	88.8	84.5	75.4	63.7	57.3	
	10% MCWB	51.1	52.2	56.9	61.7	68.0	73.7	75.3	74.1	70.5	64.0	56.4	51.6	
	0.4% WB	62.2	63.9	66.2	70.1	74.7	78.8	79.6	80.0	76.8	73.3	66.6	65.0	
	0.4% MCDB	66.4	68.8	75.9	79.9	84.1	89.2	89.9	91.6	86.7	83.4	72.3	69.6	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	59.4	61.7	63.7	67.4	72.9	77.6	78.4	78.1	75.1	71.1	63.5	61.5	
	2% MCDB	63.2	69.2	73.2	76.7	82.6	88.0	88.8	88.6	85.4	79.7	67.9	65.6	
	5% WB	56.7	57.9	61.5	65.5	71.4	76.3	77.6	76.9	73.7	68.6	60.2	57.7	
	5% MCDB	60.8	62.8	68.8	74.2	80.9	86.1	88.0	86.7	83.7	75.8	65.8	61.5	
	10% WB	52.1	53.2	58.8	63.5	69.9	75.0	76.6	75.7	72.3	66.1	57.0	52.6	
	10% MCDB	56.4	58.3	66.5	72.0	78.4	84.3	86.1	84.5	81.5	73.0	62.4	56.5	
Mean Daily Temperature Range	MDBR	17.9	18.3	20.7	22.0	20.0	20.2	19.1	20.6	21.9	22.2	20.7	16.9	
	5% DB MCDBR	21.5	23.1	25.2	25.3	22.6	22.2	21.3	24.2	25.0	26.5	25.4	20.6	
	5% DB MCWBR	16.8	16.2	14.6	12.5	9.9	8.7	8.1	8.7	9.3	12.3	15.5	15.8	
	5% WB MCDBR	19.8	20.0	20.7	19.8	19.0	19.0	18.7	19.3	20.2	20.9	19.5	18.5	
	5% WB MCWBR	17.2	15.8	13.2	11.6	9.1	8.5	8.0	8.3	8.6	11.3	14.4	16.0	
Clear-Sky Solar Irradiance	taub	0.316	0.323	0.372	0.400	0.441	0.460	0.497	0.499	0.423	0.366	0.330	0.309	
	taud	2.479	2.439	2.346	2.267	2.203	2.176	2.093	2.076	2.280	2.424	2.479	2.506	
	Ebn at Noon	279	290	283	280	269	263	252	248	263	270	271	273	
	Edh at Noon	26	31	37	42	46	47	51	50	39	31	26	24	
All-Sky Solar Radiation	RadAvg	689	874	1217	1589	1805	2004	1920	1763	1484	1135	803	585	
	RadStd	61	112	99	103	155	136	111	84	106	128	83	57	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+191	+137

Nomenclature: See separate page